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Investigations of partonic energy loss and initial state effects in small systems
via dijet measurements with the ATLAS detector

by

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Dissertation

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Abstract

This thesis presents two experimental studies of QCD using data recorded by the ATLAS experiment at the Large Hadron Collider. The first study investigates proton-size fluctuations in proton–lead collisions. The internal momentum configuration of a proton can modify its effective transverse size, particularly when a single parton carries an unusually large fraction of the proton’s momentum. Dijet production provides a means of probing this configuration because the dijet kinematics constrain the momentum fraction extracted from the proton. The relationship between this momentum fraction and both bulk particle production and forward neutron production is measured. These observables provide information about proton-size fluctuations and enable comparisons with theoretical models of the proton’s internal structure. They also provide important constraints for interpreting proton–lead collisions as a control system for heavy-ion measurements. The second study examines jet quenching in light-ion collisions using the dijet balance observable, defined as the ratio of the transverse momentum of the lower-momentum jet to that of the higher-momentum jet. Interactions between jets and the QGP modify this distribution by reducing the balance between the two jets. Measurements in oxygen–oxygen and neon–neon collisions provide access to a collision system smaller than previously studied heavy-ion systems while retaining a substantial transverse overlap between the colliding nuclei. A rigorous statistical analysis establishes that the dijet balance is significantly modified in light-ion collisions relative to proton–proton collisions. Together, these studies use dijets to investigate both the structure of cold QCD matter and the energy loss of hard-scattered partons traversing the QGP.

Acknowledgements

If you do not want to include an acknowledgment, delete the text of the acknowledgment and the `\acknowledgements` command from your `.tex` file.

A dedication is optional.

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List of Abbreviations

Abbreviation	Definition
ATLAS	A Toroidal LHC ApparatuS
LHC	Large Hadron Collider
QCD	Quantum Chromodynamics
QGP	Quark-Gluon Plasma
ZDC	Zero Degree Calorimeter

List of Symbols

Symbol	Definition
$\sqrt{s_{\text{NN}}}$	Nucleon-nucleon center-of-mass energy
p_{T}	Transverse momentum
η	Pseudorapidity
ϕ	Azimuthal angle

Chapter 1 Introduction

The purpose of this thesis is to describe a set of measurements that experimentally study the strong force in two very different temperature regimes using data recorded by the ATLAS experiment at the Large Hadron Collider (LHC). Quantum Chromodynamics (QCD) is the theory of the strong force, which acts between the fundamental constituents of matter, quarks and gluons, collectively called partons. QCD is characterized by weak interactions at high energy scales and strong interactions at low energy scales. This behavior gives rise to the interesting feature of confinement, in which quarks and gluons in a vacuum are always found in a bound, color-neutral state. However, understanding QCD in a vacuum is not enough. When QCD is coupled to an external system, thermodynamic quantities such as energy density and temperature become relevant. At sufficiently high temperature and energy density, QCD predicts a transition from confined nuclear matter to a deconfined phase near $T = 160$ MeV.

The curious nature of this phase change is that, above this temperature, the confinement principle described above no longer holds, and quarks and gluons move about freely in a state known as the quark-gluon plasma (QGP). This phase change defines two separate regimes of QCD, both worthy of study. **Cold QCD** is the state of nuclear matter in which the universe has existed for the vast majority of its history. It is important to study because it is the state we most often interact with as humans. **Hot QCD**, also known as the QGP, is the state of nuclear matter that existed in the very early universe. Since quarks and gluons are not confined to bound states, the QGP provides a laboratory for studying their interactions. Over the past 50 years, experimental heavy-ion physicists have built and used massive machines capable of probing both Cold and, more recently, Hot QCD through high-energy particle collisions.

Jets are highly energetic sprays of collimated particles that originate from high-energy scatterings in particle collisions. Due to momentum conservation, they are most commonly produced in pairs through a highly energetic $2 \rightarrow 2$ scattering process between

two fundamental particles. These jet pairs form a dijet. This thesis presents two measurements that use dijets as experimental probes to study each phase independently.

Protons are composite particles made up of three valence quarks and a number of ‘sea’ quarks and gluons. When protons are given a certain amount of energy, such as in a particle accelerator, conservation of momentum ensures that the total momentum of each proton is shared among its constituent partons, but it does not ensure equal sharing between them. Constituent partons share the momentum of the parent proton through quantum-mechanical processes. The momentum configuration of the partons can affect the overall shape of the proton. When, by chance, one parton monopolizes more than its share of momentum, the color fields of the other partons are shadowed, giving the appearance of a smaller-than-average proton. Citation needed Dijets have the unique experimental ability to measure the fraction of the momentum extracted from a projectile proton in a high-energy nuclear collision. In addition, by colliding protons with a very large target ion (Pb) the size of the projectile can be studied by measuring quantities proportional to the size of the hole left in the projectile. Two such quantities, bulk particle production and forward neutron production, are studied as a function of the momentum fraction extracted from the proton. The novel measurement presented in this thesis allows detailed comparisons with theoretical models that attempt to capture this effect and provides crucial input for the heavy-ion community, which uses proton–Pb collisions as a control case for Pb–Pb collisions.

Because dijets originate from $2 \rightarrow 2$ scattering processes, conservation of momentum requires each jet in the pair to have a similar energy. The observable x_J represents the ratio of the lower-momentum jet to the higher-momentum jet and is called the dijet balance. The distribution of x_J is known to be modified by the presence of final-state effects, i.e. the presence of a QGP medium surrounding the dijet. citation needed The stochastic manner in which jet-medium interactions occur produces dijets that are less balanced on average. New light-ion data (O+O and Ne+Ne) provide a laboratory smaller

than previous heavy-ion systems (Pb+Pb or Au+Au). Additionally, unlike asymmetric systems (p +Pb or d +Au), light ions have a sizeable transverse overlap. Results presented in this thesis take a well established observable and apply it to a new collision system. For the first time, a rigorous statistical analysis is presented establishing that the dijet balance is significantly modified in light-ion collisions with respect to pp . They also help inform models that constrain how jets lose energy to the QGP.

The thesis is organized as follows. First, the relevant theoretical background of QCD, nuclear structure, jets, and heavy-ion physics is established in [Chapter 2](#). [Chapter 3](#) discusses the LHC accelerator complex and the ATLAS experiment as well as details some technical work done by the author of this thesis. In [Chapter 4](#), the analysis of Cold QCD proton-size fluctuations in p +Pb collisions is presented and the results are discussed. Then, the observation of jet energy loss in light-ion collisions via a dijet-balance measurement is reported in [Chapter 5](#). Finally, the key results of this thesis and their implications for the field are summarized in [Chapter 6](#).

Chapter 2 Background

2.1 Quantum Chromodynamics

Quantum Chromodynamics (QCD) has been tested experimentally in many different collision systems and at many different energies. It is a field theory describing the interaction between quarks, the fermionic building blocks of matter, and gluons, the bosonic force carriers. This section introduces the basic features of QCD that are relevant background for the investigations presented in this thesis.

Gell-Mann successfully described the observed spectrum of hadrons using different flavors of *quarks* with different fractional charges [1]. This quark model was highly successful, but it predicted fermions that appeared to violate the spin-statistics theorem, such as the Δ^{++} baryon, which contains three u quarks with parallel spins [2]. This problem was resolved by introducing an additional quantum number called color charge, consisting of three color degrees of freedom, conventionally denoted r , g , and b . QCD describes the observed hadron spectrum in terms of colorless bound states. Mesons are colorless because they consist of a color–anticolor pair, while baryons are colorless because they contain all three colors or all three anticolors.

2.1.1 QCD Lagrangian

QCD is a gauge theory based on a local $SU(3)$ color charge symmetry. Just as the $U(1)$ symmetry of Quantum Electrodynamics (QED) implies the existence of a gauge field, the photon, which couples to electrically charged fermions, the $SU(3)$ symmetry of QCD implies the existence of gauge fields, the gluons, which couple to colored fermions (quarks) [3]. The elements of $SU(3)$ do not necessarily commute with one another; therefore, the $SU(3)$ symmetry is non-Abelian. The non-Abelian nature of QCD allows gluons to self-interact, unlike photons in QED.

The QCD Lagrangian is given in Equation 1 [4].

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4}F^{a\mu\nu}F_{\mu\nu}^a + \sum_f \bar{\Psi}_f (i\gamma^\mu D_\mu - m_f) \Psi_f, \quad (1)$$

D_μ is the gauge-covariant derivative, defined by

$$D_\mu = \partial_\mu - ig_s T^a G_\mu^a. \quad (2)$$

Ψ_f is the quark field of flavor f , m_f is its mass, T^a is a generator of the $SU(3)$ color group, G_μ^a is the gluon field, and g_s is the QCD coupling constant. The gluon field-strength tensor is

$$F_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c, \quad (3)$$

where f^{abc} are the structure constants of $SU(3)$. The first term in Equation 1 describes the gluon kinetic term and gluon self-interactions, while the second term describes quark propagation, quark–gluon interactions, and the quark mass term. The index f runs over the active quark flavors, and the index a runs over the eight generators of $SU(3)$. The eight gluon fields transform according to the adjoint, or octet, representation of the color group.

2.1.2 α_s , asymptotic freedom, and perturbative QCD

The QCD coupling constant g_s from Equation 1 can be used to define $\alpha_s = g_s^2/4\pi$, which is the QCD analogue of the fine-structure constant. However, the effective coupling of QCD depends not only on quark–gluon and gluon–gluon interactions, but also on higher-order diagrams involving interaction loops, which must be accounted for through renormalization. The renormalization scale, μ_R , sets the scale at which divergences are subtracted. As a result, α_s depends on the momentum-transfer scale of the QCD

interaction, Q , according to Equation 4 [2]. At one-loop order, the running coupling is

$$\alpha_s(Q) = \frac{4\pi}{\beta_0 \ln(Q^2/\Lambda_{\text{QCD}}^2)}, \quad \beta_0 = 11 - \frac{2}{3}n_f. \quad (4)$$

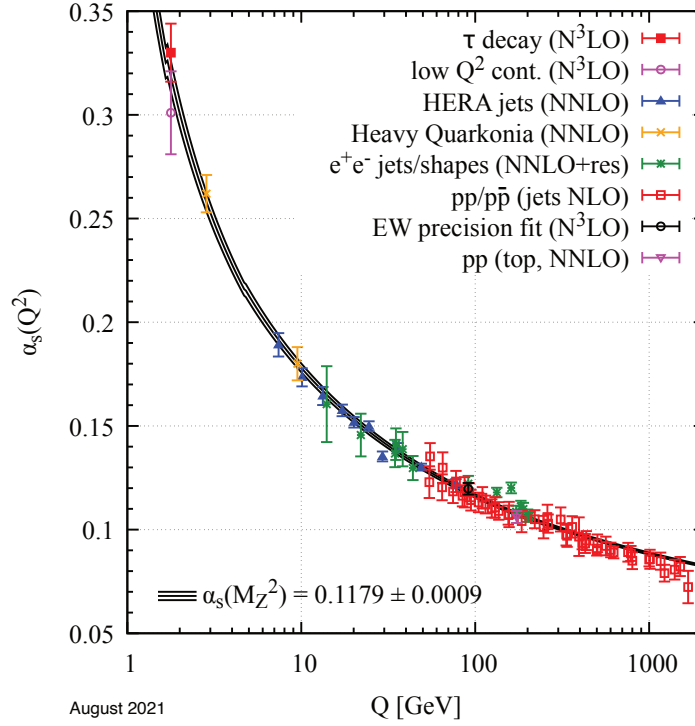


Figure 1: The running of the strong coupling α_s as a function of the momentum-transfer scale, illustrating asymptotic freedom. Taken from Reference [5]

Here, n_f is the number of quark flavors active at the scale Q , and Λ_{QCD} is the characteristic energy scale of QCD, approximately 200 MeV, depending on the renormalization scale (μ_R). For momentum transfers well above Λ_{QCD} , α_s is small and the QCD coupling becomes weak. This phenomenon is known as *asymptotic freedom* and is shown in Figure 1. In the asymptotically free regime, QCD calculations become directly accessible through perturbation theory, where observables can be estimated as a series in α_s . Perturbative QCD (pQCD) allows for highly precise predictions of many observables measured at high-energy particle colliders. Conversely, for collisions outside the perturbative regime, corresponding to low momentum transfer, calculating

observables becomes much more difficult. Sophisticated theoretical approaches such as lattice QCD can perform these calculations, but they are computationally expensive.

2.1.3 Confinement and Hadronization

The observed hadron spectrum consists of colorless (color singlet) QCD bound states. In fact, no colored bound state has ever been observed, a principle known as *confinement*. The principle of confinement can be illustrated by trying to pull a quark–antiquark pair apart. As the $q\bar{q}$ pair is separated, gluon field lines appear between the two particles, similar to the field lines between an electric dipole. Self-interactions between the gluon field lines keep them close together, creating a tube of color field between the two quarks. Figure 2 illustrates this process by comparing the gluon self-interactions in a quark–antiquark system with the field lines of an electric dipole and the compressed field lines of a color dipole.

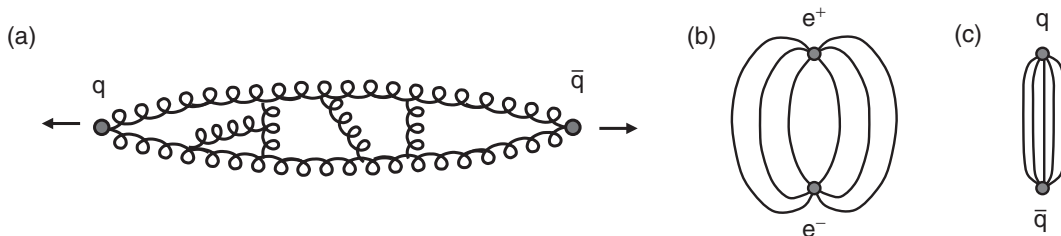


Figure 2: (left) Illustration of gluon self-interactions in a $q\bar{q}$ system. (middle) Field lines for an electric dipole. (right) Compressed field lines for a color dipole. Taken from Reference [3].

As the separation increases, this tube becomes longer; therefore, the energy required to maintain this configuration is linearly proportional to the distance. The energy density of this tube is experimentally found to be approximately 1 GeV/fm, which explains why it is energetically favorable for colored objects to form colorless states [3].

Now that the principle of confinement has been established, consider the situation in which each (anti-)quark in a $q\bar{q}$ pair is given a large momentum that pulls the pair apart, as in sufficiently hard $e^+e^- \rightarrow q\bar{q}$ scattering. As described above, the $q\bar{q}$ pair forms a tube

of gluon flux with highly concentrated energy. At some point, the energy of the gluon flux becomes large enough to produce new $q\bar{q}$ pairs, which in turn break the original flux tube into two smaller flux tubes, or ‘strings.’ The original highly energetic quarks lose some energy in this breaking process because energy is required to produce the new strings. This string lengthening and binary breaking process will repeat itself until all the $q\bar{q}$ pairs are formed into colorless hadrons. The result is two streams of highly collimated hadrons, one in the direction of the original high-energy quark and the other in the direction of the high-energy antiquark. The formation of these hadron streams, which is what experimentalists detect, is called *hadronization* and is illustrated in Figure 3 [3].

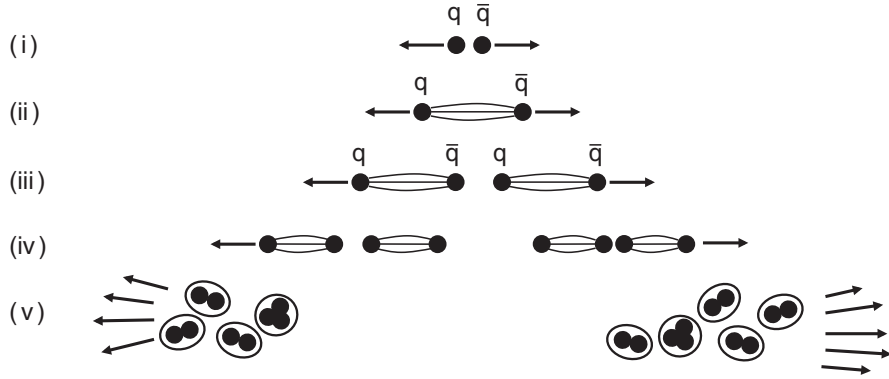


Figure 3: Schematic illustration of hadronization through the breaking of a color flux tube into quark–antiquark pairs and collimated streams of hadrons. Taken from Reference [3].

2.1.4 The Parton Model and Deep-Inelastic Scattering

The parton model provides a description of the internal structure of hadrons by imagining the hadron as a box of free point-like quarks and gluons, collectively called partons [6]. Since the hadron has momentum p the partons each have momentum $\xi_i p$ where $\sum_{\text{partons}} \xi_i = 1$. The partons themselves are imagined to be practically free of interactions, which requires that the hadron have a large momentum. The fast-moving nature of the hadron ensures that, in the laboratory frame, the internal dynamics between partons occur over a relatively long timescale. When an experiment collides with

the hadron, the internal structure is effectively frozen over the timescale of the collision.

The parton model was introduced by Feynman [6], but did not find acceptance until later experimental developments. High-energy lepton-nucleon scattering played a fundamental role in determining the internal configuration of nucleons. Taking $e^- + p \rightarrow e^-' + X$ scattering as an example, it is illustrative to define some kinematics using the diagram shown in Figure 4.

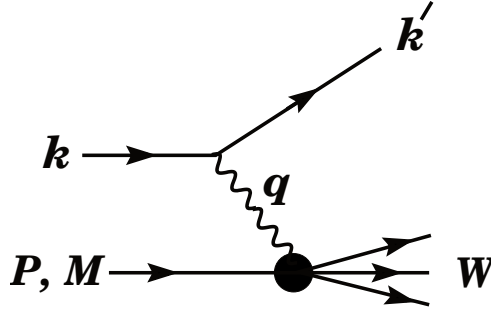


Figure 4: Schematic diagram of deep-inelastic $e^- + p$ scattering, where the electron has initial four-momentum k and the proton has mass M and four-momentum P . Taken from Reference [5].

$$Q^2 = -q^2, \quad x = \frac{Q^2}{2P \cdot q}, \quad y = \frac{q \cdot P}{k \cdot P}, \quad W^2 = (P + q)^2 \quad (5)$$

Here, q is the four-momentum transferred by the exchanged virtual photon, Q^2 is the squared momentum-transfer scale, x is the Bjorken scaling variable, y is the fractional energy loss of the electron, and W^2 is the invariant mass squared of the hadronic final state. The Bjorken scaling variable, typically called Bjorken- x , has the physical interpretation of the partonic momentum fraction (ξ_i) carried by the struck parton in the proton. The above process is called deep ($Q^2 \gg M^2$) inelastic ($W^2 \gg M^2$) scattering (DIS) [5].

Experimentally, one can measure the scattering angle (θ) and the initial (E_e) and final (E_e') energies of the electron, which can be mapped onto the above kinematic

variables using the following expressions.

$$Q^2 = 2E_e E'_e (1 + \cos \theta), \quad x = \frac{Q^2}{2M(E_e - E'_e)}, \quad y = 1 - \frac{E'_e}{E_e} \quad (6)$$

For unpolarized electromagnetic DIS in the one-photon-exchange approximation, the cross section can be written differentially in x and Q^2 as

$$\frac{d^2\sigma}{dx dQ^2} = \frac{4\pi\alpha^2}{Q^4} \left[\left(1 - y - \frac{M^2 y^2}{Q^2}\right) F_2(x, Q^2) + y^2 F_1(x, Q^2) \right] \quad (7)$$

where α is the electromagnetic fine-structure constant. F_1 and F_2 are *structure functions* that encode information about the internal charge configuration of the hadron. Bjorken predicted that, in the high-energy limit, the structure functions only depend on Bjorken- x and not on Q^2 (i.e. $F(x, Q^2) \rightarrow F(x)$) [7] This is consistent with the parton model: if the internal structure of the proton is described by point-like partons, scattering at different momentum scales does not resolve any additional structure. Confirmation that DIS involves lepton scattering from quarks inside the hadron was provided by the observation of the Callan–Gross relation [8], $F_2(x) = 2xF_1(x)$, at the Stanford Linear Accelerator Center (SLAC) [9].¹ This relation is a prediction for electron scattering from spin-1/2 constituents, whereas $F_1(x) = 0$ would indicate scattering from spin-0 constituents.

2.1.5 Factorization

While the matrix element for a given QCD process may be able to be calculated to some order in α_s , an observable will have to undergo long distances and will hadronize before it can be observed experimentally. Additionally, for collision systems with hadrons the internal partonic configuration of the reactants must be considered. Therefore, any measurement with hadrons in the initial- or final-state is sensitive to low-energy

¹Gluons cannot directly interact with the virtual photon in DIS.

(non-perturbative) QCD interactions. The running coupling of α_s in QCD can be exploited in order to derive predictions for lepton-hadron and hadron-hadron large momentum transfer cross sections. This is done by *factorization* theorems, which systematically separate (factorize) long-distance from short-distance behavior.

This is set by the factorization scale, μ_F , which partitions the calculation into the perturbative hard-scattering ($Q > \mu_F$) and the non-perturbative long range interactions ($Q < \mu_F$). Schematically, this separation is represented in Figure 5.

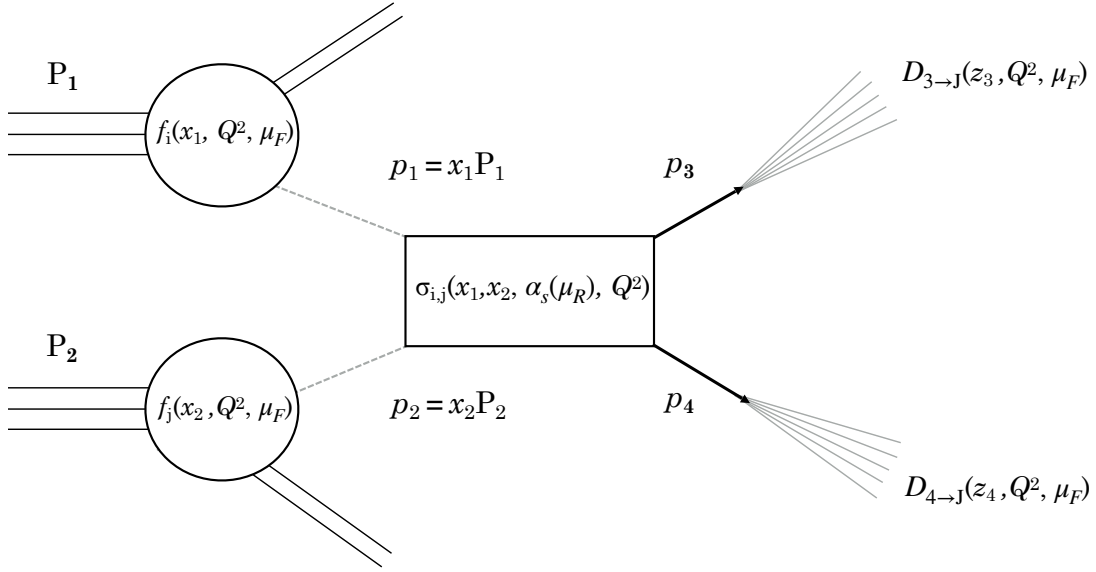


Figure 5: Schematic representation of QCD factorization.

For example, the factorized cross section for producing a dijet system in a proton-proton collision can be written schematically as

$$d\sigma_{pp \rightarrow JJ} \approx \sum_{i,j,k,l} \int dx_1 \int dx_2 \int dz_3 \int dz_4 f_i(x_1, \mu_F^2) f_j(x_2, \mu_F^2) \otimes d\hat{\sigma}_{ij \rightarrow kl} \otimes D_{k \rightarrow J}(z_3, \mu_F^2) D_{l \rightarrow J}(z_4, \mu_F^2). \quad (8)$$

Here, f_i and f_j are the initial-state parton distribution functions, $d\hat{\sigma}_{ij \rightarrow kl}$ is the perturbatively calculable hard-scattering cross section, and $D_{k \rightarrow J}$ and $D_{l \rightarrow J}$ describe the fragmentation of the outgoing partons into the observed jets. The momentum fractions

x_1 and x_2 describe the initial-state partons, while z_3 and z_4 describe the momentum fractions carried by the observed jets.

2.2 Parton Distribution Functions

2.3 Nuclear Structure

2.3.1 Accelerator considerations for nuclear species

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2.4 Small to Large Collision Systems

2.4.1 Geometry of a nuclear collision

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2.4.2 Cold nuclear matter effects

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2.4.3 Color transparency (GPD) in p+A collisions

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2.5 Quark Gluon Plasma

2.5.1 Soft probes of the QGP

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2.5.2 Hard probes of the QGP

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2.6 Jets

2.6.1 Jets

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2.6.2 Jets in p+A collisions

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2.6.3 Theoretical predictions

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2.6.4 Measurements from other collaborations

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Chapter 3 Experimental Apparatus

3.1 the LHC

In addition to its flagship high luminosity pp program the LHC also provides special runs aimed specifically for the field of heavy ion physics. A list of these runs are compiled in Table 1. These include heavy ion runs (A+A) to study QGP physics, small systems runs (p +Pb and $A_{\text{light}} + A_{\text{light}}$) to study cold nuclear matter effects and search for the turn on of QGP physics, and pp reference runs as a comparison point for the other systems at the same energy. The author of this thesis was on site assisting with operations and data taking for all Run 3 runs sans the 2026 Pb+Pb run.

Year(s)	Run	Energy ($\sqrt{s_{\text{NN}}}$)	Species	Luminosity
2010	Run 1	2.76 TeV	Pb+Pb	$\sim 10 \mu\text{b}^{-1}$
2011	Run 1	2.76 TeV	Pb+Pb	$\sim 0.15 \text{ nb}^{-1}$
2013	Run 1	2.76 TeV	pp (ref.)	$\sim 5 \text{ pb}^{-1}$
2013	Run 1	5.02 TeV	p +Pb	$\sim 30 \text{ nb}^{-1}$
2015	Run 2	5.02 TeV	Pb+Pb	$\sim 0.5 \text{ nb}^{-1}$
2015	Run 2	5.02 TeV	pp (ref.)	$\sim 28 \text{ pb}^{-1}$
2016	Run 2	5.02 TeV	p +Pb	$\sim 0.5 \text{ nb}^{-1}$
2016	Run 2	8.16 TeV	p +Pb	160 nb^{-1}
2017	Run 2	5.44 TeV	Xe+Xe	$\sim 0.3 \mu\text{b}^{-1}$
2017	Run 2	5.02 TeV	pp (ref.)	$\sim 270 \text{ pb}^{-1}$
2018	Run 2	5.02 TeV	Pb+Pb	$\sim 1.8 \text{ nb}^{-1}$
2023	Run 3	5.36 TeV	Pb+Pb	$\sim 1.8 \text{ nb}^{-1}$
2023	Run 3	5.36 TeV	pp (ref.)	$\sim 400 \text{ pb}^{-1}$
2024	Run 3	5.36 TeV	Pb+Pb	$\sim 1.7 \text{ nb}^{-1}$
2024	Run 3	5.36 TeV	O+O	$\sim 8 \text{ nb}^{-1}$
2024	Run 3	5.36 TeV	Ne+Ne	$\sim 1 \text{ nb}^{-1}$
2025	Run 3	5.36 TeV	Pb+Pb	$\sim 2.8 \text{ nb}^{-1}$
2026	Run 3	5.36 TeV	Pb+Pb	$\sim 2.8 \text{ nb}^{-1}$

Table 1: Summary of nuclear physics data-taking runs at the Large Hadron Collider (LHC), detailing runs for nuclear physics.

3.1.1 2016 p +Pb Data

In 2016, the LHC collided the asymmetric proton–Lead (p +Pb) system. Two different center of mass energies were used, $\sqrt{s_{\text{NN}}} = 5.02$ TeV and $\sqrt{s_{\text{NN}}} = 8.16$ TeV. The luminosity sampled by the higher energy sample far exceeded that of the lower energy sample. Part of the reason for that discrepancy comes from the difference in the average number of collisions per bunch crossing ($\langle\mu\rangle$) in each sample is much higher in the $\sqrt{s_{\text{NN}}} = 8.16$ TeV sample. Higher $\langle\mu\rangle$ also means higher level of contamination from in-time and out-of-time pileup events.

Another particular feature of the 2016 p +Pb dataset is the fact that the direction of circulation for the proton and lead beams switched in the middle of the run. In the first hi- $\langle\mu\rangle$ period, **Period B**, protons circulated in the accelerator in a clockwise direction² while lead ions circulated in the counterclockwise direction. In the second hi- $\langle\mu\rangle$ period, **Period C**, this directionality was reversed.

3.1.2 2025 Light Ion Data

3.2 ATLAS detector

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3.3 Zero Degree Calorimeter

The ATLAS Zero Degree Calorimeter (ZDC) is designed to detect neutral particles that are very close to the beampipe. It consists of two detectors located in absorbers along the beamline ± 140 m from the ATLAS interaction point. Each side (or arm) consists of four modules each with a material budget of 1.14 hadronic interaction lengths. The sampling calorimeter design consists of interspersed absorber layers, which cause the incident particles to shower, and active layers, which output a signal proportional to the energy deposited by charged particle shower crossing that layer. The absorber layers consist of

²direction defined from viewing the LHC from above

tungstan plates, and the active layers consist of cherenkov radiators. These Cherenkov radiators transport Cherenkov radiation³ induced by the particle shower to the lens of a Photomultiplier tube (PMT). Each of the four modules are instrumented with a single PMT. Cherenkov radiators were chosen since they offer substantial radiation hardness, quite necessary for a location that receives a very high dose. In Run 2, the active layers consisted of Quartz rods as the active layer. The radiation hardness of the detector was improved considerably in Run 3 when the active layer was replaced with doped fused silica rods.

As a calorimeter, the ZDC destructively samples the energy of the particles that enter it. Since dipole magnets between the IP and the ZDC swipe away forward charged particles, only neutral particles enter the ZDC. In ion running, the fragmentation of the ion's nucleus results in a population of neutrons interacting in the ZDC. Since these so called "spectator" neutrons do not interact in the collision at the IP they will impact the ZDC with roughly the same per nucleon energy as the beam. The number of spectator neutrons is non-trivially dependent on the event geometry, but they range from zero to roughly $(A - Z)/2$.

In order to capture signals from a zero to many neutrons, the ZDC operates with two sets of readout instrumentation, denoted as high and low gain readouts. The high gain readout is configured to have a dynamic range that saturates with a couple of neutrons. The low gain readout has a range of a roughly 10x that of the high gain readout.

The ZDC reconstruction software transforms the raw data read out from the above ADCs into module specific amplitudes. Firstly, each waveform must have its pedestal subtracted in order to make sure that the output amplitude represents only strength of the module signal and not any offset caused by the electronics configuration. Then, the subtracted samples can be fit to a functional waveform. The amplitude that minimizes the χ^2 of this fit is then considered to be the amplitude in that given event. **Prob need to**

³Cherenkov radiation is emitted in a Mach cone when particles travel faster than the local speed of light in the material medium

include more details about the ZDC reconstruction

Finally, the ZDC calibration transforms the above amplitudes, which have units of ADC counts, into units of energy.

Missing ZDC Module in 2016 Data During the 2016 p +Pb run, the ZDC side C module 0 (EM) was removed and replaced with the Arm2 calorimeter of the LHCf experiment [10]. Figure 6 displays the transverse acceptance of both the ATLAS ZDC and LHCf. The ATLAS ZDC EM module is the first of four sequential calorimeter modules, counting from the perspective of looking at the ZDCs from the ATLAS interaction point; therefore, the LHCf calorimeter would see forward neutral products before they reach the side C ZDC. The LHCf detector was put in place to measure the forward neutral products of the proton-going side in the **Period B** orientation; however, in the **Period C** orientation the LHCf detector was retracted into the “garage” position to avoid radiation damage [10]. The “garage” position indicates an upward vertical shift of 112 mm from the nominal position shown in Figure 6. This shift places the full LHCf Arm2 calorimeter outside of the full acceptance of the ATLAS ZDC. The side C ZDC has roughly 75% of the material budget of side A. This effectively reduces the interaction length of the side C ZDC from 4.6 to 3.4, which in turn reduces the neutrons expected to shower in the ZDC from 99% to 96.6%.

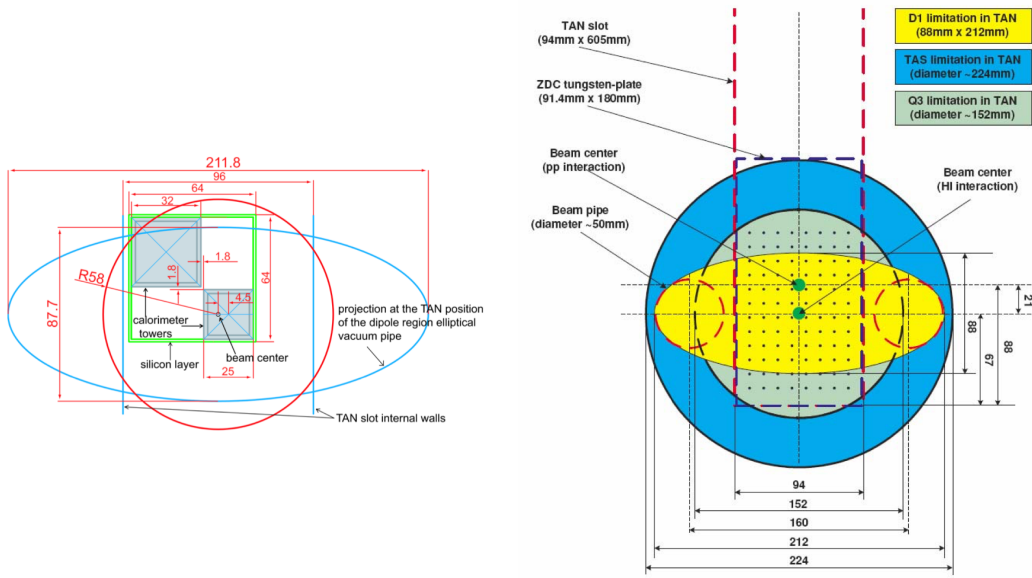


Figure 6: Detector geometry for LHCf Arm2 [11] (left) and the ATLAS ZDC [12] (right), viewed from the ATLAS interaction point. Distances are given in mm. The blue oval in the left image corresponds to the boundary of the yellow oval in the right image. The active area of LHCf is represented by the filled blue squares, and the active area of the ZDC is bounded by the purple dashed line.

3.3.1 2016 p +Pb ZDC Calibration

Due to the asymmetric nature of the p +Pb collisions system in each period only one arm of the ZDC detector will receive ion fragments.

In 2019, J. Nagle performed [ref needed](#) the calibration of the 5.02 TeV data, but concluded that the contamination from pileup in the high- $\langle\mu\rangle$ p +Pb data made a calibration unfeasible.

The author of this thesis undertook the calibration of the high- $\langle\mu\rangle$ p +Pb data in 2023 after substantial improvements had been made to the ZDC reconstruction in order to mitigate the previously untenable issues.

The first improvement made to the ZDC reconstruction comes from [\[13\]](#). This note documents updates to the signal processing created mainly for the 2018 Pb+Pb data, but that are useful for the 2016 p +Pb data as well. These updates include significant improvements to the reconstruction's ability to fit pulses in the presence of out of time (OOT) pile up, both preceding and after the event in question. Large OOT pileup pulses would saturate the high gain ADC, leading to the pulse being analyzed with the low gain ADC which has a roughly 10x worse resolution. By excluding a few samples from the beginning or (end) of these pulses and looking at the high gain ADC values, the reconstruction will yield a reasonable fit for the current event. This is especially important given that the relatively high $\langle\mu\rangle$ in the 2016 8.16 TeV p +Pb has more events affected by out of time pileup.

The second significant improvement made to the ZDC reconstruction was the ‘two-pass’ analysis method. This method first uses a relatively high second derivative threshold to search for pulses in a ZDC event. This high threshold helps to suppress noise from small background and pileup pulses. However, if at least one module has a pulse that passes this threshold, all the modules on said ZDC arm are reanalyzed with a lower second derivative threshold. This ‘repass’ recovers finer pulses that were too small to be ‘seen’ on the first pass.

The third significant improvement⁴, involves the studies on the pulse timing.

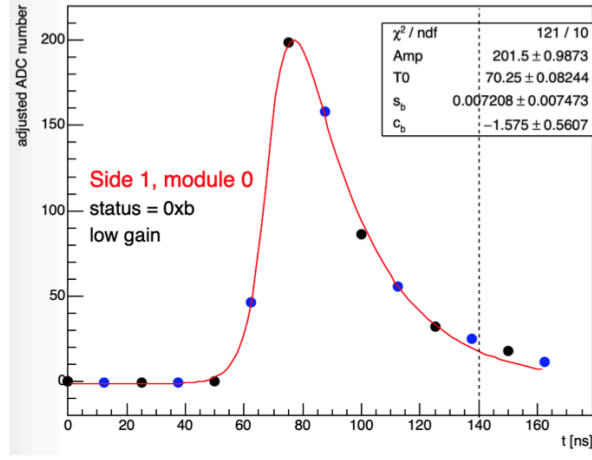


Figure 7: Example of a fitted ZDC pulse, taken from [13]

Base ZDC pulses (see Figure 7) are fit to a function of the form given in Equation 9, hereafter referred to as a FermiExp pulse.

$$f(t | A, \tau_1, \tau_2, t_0, C) = A \left(\frac{e^{-(t-t_0)/\tau_2}}{1 + e^{-(t-t_0)/\tau_1}} \right) + C \quad (9)$$

Several of these parameters are fixed in order for the fitting function to have fewer degrees of freedom and thus converge more efficiently. For example, τ_1 & τ_2 are fixed in the 2016 ZDC reconstruction.

From a data quality standpoint, it would be useful to be able to reject pulses that were relatively early or late when compared to other pulses. One way to do this would be by defining a nominal t_0 range that is acceptable. However, there are two main challenges to defining a nominal t_0 .

1. When the high voltage powering the a ZDC PMT changes, it affects the transit time in the tube which leads to a slightly different nominal t_0 .

Solution: Implement a time dependent nominal t_0 correction for the entire dataset.

⁴This improvement was done by the author of the thesis, while the first two were done by Pengqi Yin of Columbia University

2. There is a nonlinear dependence of the t_0 on the pulse amplitude that must be corrected.

Solution: Fit the curvature induced by the nonlinear dependence to a 5th order polynomial. Use that fit function to flatten the convex distribution.

The data used for the calibration of the high- $\langle\mu\rangle$ p +Pb was read out into a dedicated ZDC calibration stream. This dedicated readout only stores information about the ZDC detector for each accepted event⁵ and can therefore accumulate substantial statistics needed for a precision calibration. Events entering the stream fired the **L1_ZDC_A** or **L1_ZDC_C** triggers, requiring that there was some substantial energy in at least one arm of the ZDC. To get a set of events unbiased by trigger turn on thresholds, the set of events used to analyze Side A were triggered on a signal in Side C and vice versa.

The ZDC is designed to sample neutral ion fragments with the same per-nucleon energy as the beam. The lightest neutral fragment possible is a single beam energy neutron incident on the detector. A single neutron that fragmented from a lead ion will, on average, be as energetic as the per-nucleon beam energy, which here is 2.5625 TeV. Therefore, the energy of single neutrons gives a well defined energy to calibrate against. Of course single neutrons will not have the exact per-nucleon beam energy every time, since the exact energy of bound nucleons is stochastically modified by the so-called ‘fermi-momentum’ **citation needed**. The variance of single neutrons given by fermi-momentum coupled with the experimental resolution of the ZDC should give the width of the detector response to a single neutron.

Therefore the single neutron peak⁶ intrinsically contains information about the ZDC Energy Scale (μ_{1n}) and ZDC Energy Resolution (σ_{1n}). However, we are interested in not just any calibration, but the *optimal* one. The optimal calibration is defined to be the

⁵The per-event data footprint of the ZDC detector is roughly an order of magnitude smaller than the whole ATLAS detector including all detector subsystems

⁶observed detector response to single neutrons

one that puts the mean of the single neutron peak at 2.5625 TeV and minimizes the variance (therefore maximizing experimental resolution). The machinery of the calibraton works as follows.

The goal is to find a set of gains $\mathbf{g} = \{g_0, g_1, g_2, g_3\}$ such that given the ZDC module amplitudes for a given event $\mathbf{A} = \{A_0, A_1, A_2, A_3\}$ the dot product $\mathbf{g}^T \mathbf{A}$ is equal to the energy of the ZDC in units of GeV(2562.5 GeV). The below calibration procedure is performed on a subset of ZDC events in the first neutron peak. Equations 10 and 11 are the mean an variance of the first neutron peak sample. The variable N indicates the size of the sample used to perform the analysis.

$$E_{1n} = \langle \mathbf{g}^T \mathbf{A} \rangle_N \quad (10)$$

$$\sigma^2 = \langle (\mathbf{g}^T \mathbf{A})^2 \rangle_N - (\langle \mathbf{g}^T \mathbf{A} \rangle_N)^2 = \langle (\mathbf{g}^T \mathbf{A})^2 \rangle_N - (E_{1n})^2 \quad (11)$$

Using the method of Lagrange multipliers, one can minimize the variance under the constraint that mean of the one neutron peak is equal to the per nucleon beam energy.

$$\mathcal{L} = \langle (\mathbf{g}^T \mathbf{A})^2 \rangle_N - (E_{1n})^2 + \lambda (\langle \mathbf{g}^T \mathbf{A} \rangle_N - E_{1n}) \quad (12)$$

Solving 12 with respect to $\mathbf{g}$ and λ yields the following system of equations.

$$\begin{bmatrix} 2 \langle A_0 A_0 \rangle & 2 \langle A_0 A_1 \rangle & 2 \langle A_0 A_2 \rangle & 2 \langle A_0 A_3 \rangle & \langle A_0 \rangle \\ 2 \langle A_1 A_0 \rangle & 2 \langle A_1 A_1 \rangle & 2 \langle A_1 A_2 \rangle & 2 \langle A_1 A_3 \rangle & \langle A_1 \rangle \\ 2 \langle A_2 A_0 \rangle & 2 \langle A_2 A_1 \rangle & 2 \langle A_2 A_2 \rangle & 2 \langle A_2 A_3 \rangle & \langle A_2 \rangle \\ 2 \langle A_3 A_0 \rangle & 2 \langle A_3 A_1 \rangle & 2 \langle A_3 A_2 \rangle & 2 \langle A_3 A_3 \rangle & \langle A_3 \rangle \\ \langle A_0 \rangle & \langle A_1 \rangle & \langle A_2 \rangle & \langle A_3 \rangle & 0 \end{bmatrix} \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ g_3 \\ \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ E_{1n} \end{bmatrix} \quad (13)$$

The ZDC energy response changes significantly over the period of a run. Therefore, it is imperative to create a time dependent calibration.

The software to implement this calibration was created by the author of this thesis. Some important practical details are described below. Firstly, the data is broken a series of separate consecutive time intervals with the condition that each time interval has about a million events, which was found to be sufficient for a stable calibration. Next, within each interval, a histogram is created out of the sum of ZDC module amplitudes for a given event $\Sigma_i^4 A_i$. The single neutron peak is apperent in the spectrum of $\Sigma_i^4 A_i$, and a window $\pm 1.5\sigma$ around the peak is defined by a simple Gaussian fit. Then, for each time interval, fourteen histograms are defined corresponding to the unaveraged matrix elements of Equation 13 and are filled with events falling inside the single neutron window defined in the previous step. Next the histograms are averaged over and Equation 13 can be solved for the optimal $\mathbf{g}$ for each time interval. Once the optimal $\mathbf{g}$ for every time interval is decided, a cubic spline is used to interpolate smoothly between the sucesive time intervals. Then, on an event by event basis the calibrated ZDC energy is equal to $\mathbf{g}(\mathbf{t})^T \mathbf{A}$.

The optimized gains as a function of time ($\mathbf{g}(t)$) for Period B of the p +Pb run is displayed in Figure 8. Citation needed Each module is denoted by a different color, but the overall trend is common to all modules. Clusters of points on the time-axis represent individial LHC running periods. Throughout individual running periods, the gains increase. This increase is caused by accumulated radiation damage to the detector, which lowers the average module amplitudes ($\mathbf{A}$) requiering higher gain factors ($\mathbf{g}$) to calibrate to the same energy. Running periods with higher levels of detector degriation (higher $\frac{d\mathbf{g}}{dt}$) corruspond to periods of data taking with higher- $\langle\mu\rangle$ and thus more incident radiation and integrated luminosity. The gain increses in Figure 8 are not monotonic as expected from continous radiation damage. This is because at certain points throught the run, the operating power supply for the module PMTs were manually adjusted⁷ to elicit a stronger detector response. Each of these changes was correlted with 2016 logbook

⁷The high voltage of the ZDC PMTs was increased

entires from Peter Steinberg. Additionally, the data quality recommendations had to be adjusted to make sure that no data was used when the detector configuration was actively being changed.

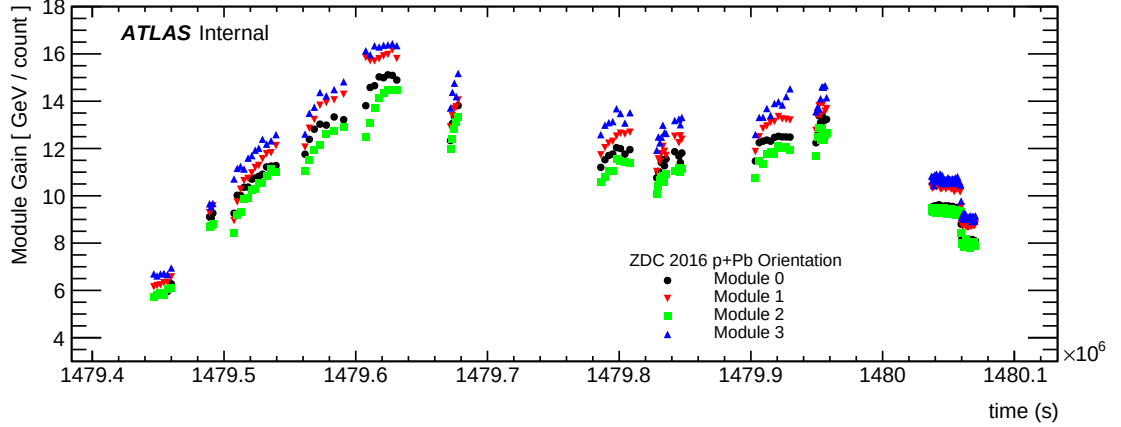


Figure 8: Citation and description needed

An example of the ZDC energy spectrum and single neutron peak is displayed in Figure 9. Following calibration, a simultaneous fit is performed to the first three neutron peaks, as well as a background term. Background sources include neutral pions, photons, and beam–beam backgrounds. The functional form of the background term is chosen to facilitate fit convergence in the 1–3 neutron range [14].

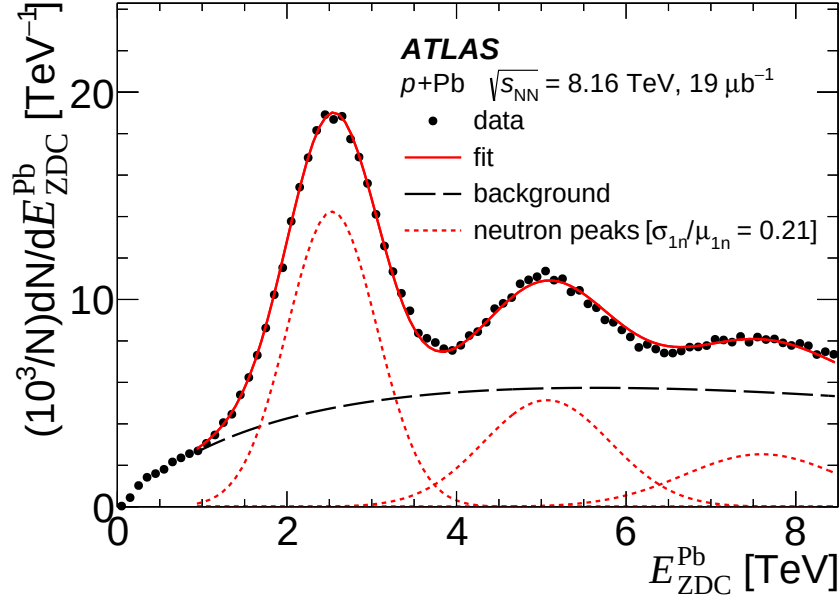


Figure 9: ZDC background estimation for run 313100, taken from [14].

3.4 ATLAS Jet Reconstruction

When studying HI collisions with the ATLAS detector, jets are reconstructed using the anti- k_t jet clustering algorithm [15] with calorimeter towers as constituents. This algorithm uses collections of four-vectors corresponding to a set of tracks of charged particles in the inner tracking system (aka “track jets”), massless calorimeter towers with size $\Delta\eta \times \Delta\phi = 0.1 \times 0.1$ (aka “heavy ion jets”), or sets of simulated stable particles at the generator level (aka “truth jets”). The measurements presented in this thesis make use of heavy ion jets reconstructed using the anti- k_t algorithm with $R = 0.4$, called **AntiKt4HIJets**.

A detailed review of the performance of jet reconstruction in heavy ion collisions can be found in [16]. This note will only discuss the main features of the heavy ion jet reconstruction and their implications on the dijet balance measurement.

3.4.1 Underlying Event Subtraction

To reconstruct jets in HI collisions, a background arising from the underlying event (UE) must be subtracted from each tower to avoid significant rates of fake jets [17] and to avoid distortion of the jet energy scale. The energy registered in each cell can be decomposed into contributions from a jet and from the UE as

$$\frac{d^2 E_T^{\text{total}}}{d\eta d\phi} = \frac{d^2 E_T^{\text{jet}}}{d\eta d\phi} + \frac{d^2 E_T^{\text{UE}}}{d\eta d\phi} \quad (14)$$

Only the first contribution should be used to reconstruct jets. Therefore, $\frac{d^2 E_T^{\text{UE}}}{d\eta d\phi}$ must be estimated and subtracted. The UE subtraction procedure makes use of the UE average transverse energy density, $\rho(\eta, \phi)$. The ϕ dependence in ρ arises from global azimuthal correlations in the particle production, usually referred to as flow, that arises from the hydrodynamic response of the medium to the geometry of the initial collision [18]. One can describe the UE contribution to the transverse energy of the calorimeter towers by using the amplitudes (ν_n) and the phases (Ψ_n) of the Fourier components of the azimuthal angle distributions:

$$\rho(\eta, \phi) = \frac{1}{2\pi} \frac{\langle dE_T \rangle}{d\eta} \left(1 + 2 \sum_{n=2}^N \nu_n \cos(n(\phi - \Psi_n)) \right) \quad (15)$$

where ϕ and n represent the azimuthal angle of the tower and the order of the flow harmonic, respectively. Although ν_2 dominates, the third and the fourth harmonics are used to further improve the ρ evaluation. $N = 4$ was used to reconstruct Run 2 Pb+Pb data, while for p +Pb and pp data used for the analysis presented in this note, no anisotropic component was allowed in the fit, *i.e.* $N = 0$. This choice was made due to the relatively low multiplicity characterizing p +Pb and pp collisions compared to Pb+Pb.

The average transverse energy density is estimated using an iterative procedure that excludes regions populated by jets. Please note that, for both pp reference, p +Pb and

light ion data, the HIP reconstruction mode [19], where the HI jet reconstruction is deployed accounting for low underlying event conditions, is applied. HIP mode does not include any correction for azimuthal anisotropy of the UE.

A first estimate of the UE average transverse energy density, $\rho(\eta)$, is evaluated in 0.1 intervals of η excluding (η, ϕ) regions of the calorimeter populated by “seed” jets. During this first step, the seeds are defined as a set of $R = 0.2$ HI jets and $R = 0.4$ track jets. To be considered as a seed, an $R = 0.2$ jet has to pass the discriminant cut on the value of maximal-over-mean tower energy, e.g. $\max/\text{mean} > 4$. This cut enhances purity of the selected jet candidates against UE correlations. The background is then subtracted from the jet calorimeter towers to compute the subtracted jet kinematics. At this stage, ρ and ν_n are recalculated by excluding (η, ϕ) region of the calorimeter containing a jet seed.

3.4.2 Jet Calibration

This section describes the general features of the HI jet calibration strategy. Since the two measurements reported in this thesis were taken nine years apart, they use slightly different calibration strategies⁸.

Individual calorimeter cells are themselves calibrated to the EM scale, but there is a variety of factors (*e.g.* passive material in the detector, gaps between cells, signal losses, etc. [20]) that allow only for a partial measurement of the energy deposited by a single jet in the detector. To compensate for these experimental effects, the energy scale of reconstructed jets needs to be calibrated to the truth energy scale. The calibration sequence for HI jets is presented in [21]. Here we briefly review the main steps of this procedure. After the UE subtraction, three different types of calibrations are applied, namely:

1. origin correction to account for the position of the primary vertex in each event.

⁸Differences in the calibration strategies between the two measurements will be reported on a per measurement level

2. EtaJES MC-based calibration, to recover the truth energy scale in MC.
3. In-situ γ +jet, Z+jet and η -intercalibration, to account for differences between MC and data.

A small pseudorapidity correction is also applied to the jet reconstructed η , to counteract the implicit assumption that the z -vertex⁹ = 0 made during the reconstruction of the data. For this reason, this procedure is called “origin correction”.

The EtaJES calibration is the larger between the two and is derived from PYTHIA8 simulated dijet events. The simulations are carried out using the MC conditions and the energy characteristic of the data taking. The scope of this calibration, common to both data and MC, is to correct from the EM scale to the truth hadronic scale. The per-jet energy correction factor is computed using a numerical inversion procedure [22, 23], see [24] for a review of the procedure’s math. A typical jet energy scale correction, $E_T^{\text{true}}/E_T^{\text{reco}}$, ranges from ~ 1.2 to 2.0 .

The second calibration component is only applied to data to account for discrepancies between MC and data. The in-situ and η -intercalibrations implement residual corrections to the jet energy scale based on the p_T balance in vector boson (γ , Z^0) - jet events (the *in-situ* correction) and dijet events where mid-rapity jets are used to calibrate the forward rapidity jets (the η -intercalibration). Calibration factors are derived in bins of p_T and η and are applied as a correction to the jet p_T .

⁹Position of the vertex associated to the jet along the beam axis.

Chapter 4 Dependence of geometry estimators on hard scattering kinematics in p+Pb collisions

Just like A+A collisions, hard-process rates in p +A collisions are studied as a function of centrality, defined using global event-level observables sensitive to collision geometry. In the presence of a hard scattering, however, the proton configuration can modify the relation between those observables and the impact parameter, contaminating the above correlation. Measuring the estimators directly as a function of the reconstructed partonic kinematics inverts the usual centrality-selected approach and quantifies this bias.

This chapter presents the first measurement of event geometry estimators on hard-scattering kinematics in p +Pb collisions. Dijets are used as a proxy for the initial-state partonic scattering configuration [25] and their kinematics are correlated with the Pb-going flow forward transverse energy and zero-degree neutral energy.

4.1 Measured Observables

The measured observables are the neutral energy deposited in the Pb-going Zero Degree Calorimeter (ZDC), $E_{\text{ZDC}}^{\text{Pb-side}}$, and the transverse energy measured in the Pb-going Forward Calorimeter (FCal), E_{T}^{Pb} . The ZDC is sensitive primarily to spectator neutrons emitted during nuclear breakup, whereas the FCal transverse energy is an event-activity estimator sensitive to particle production by the collision participants. Both quantities have been used to characterize the geometry of proton–nucleus collisions.

The hard-scattering kinematics are accessed using the two highest-transverse momentum jets in each event. The momentum fraction carried by the parton in the proton is estimated at leading order as

$$x_p = \frac{p_{\text{T},1} e^{y_1^{\text{CM}}} + p_{\text{T},2} e^{y_2^{\text{CM}}}}{\sqrt{s_{\text{NN}}}}, \quad (16)$$

where $p_{\text{T},1}$ and $p_{\text{T},2}$ are the transverse momenta of the leading and subleading jets and

y_1^{CM} and y_2^{CM} are their rapidities in the nucleon–nucleon center-of-mass frame. This estimator reconstructs the Bjorken- x of the scattered proton parton from final-state jet kinematics. Simulation studies show that the reconstructed quantity is, on average, 6–8% lower than the parton-level value in the measurement phase space, with a relative resolution that improves from about 10% at low x_p to 4% at high x_p .

Normalized $E_{\text{ZDC}}^{\text{Pb-side}}$ and E_{T}^{Pb} distributions are measured in intervals of x_p , along with their means, $\langle E_{\text{ZDC}}^{\text{Pb-side}} \rangle$ and $\langle \text{FCal} \sum E_{\text{T}}^{\text{Pb-side}} \rangle$. The distributions expose changes in shape, while the means provide compact measures of the x_p dependence. The correlation between $E_{\text{ZDC}}^{\text{Pb-side}}$ and E_{T}^{Pb} is also studied in each x_p interval to connect the spectator-neutron and participant-activity observables.

4.2 Data Sets

4.2.1 2016 $p+\text{Pb}$ Data

The $p+\text{Pb}$ data at $\sqrt{s_{\text{NN}}} = 8.16$ TeV used for this analysis were delivered by the LHC to ATLAS in November and December of 2016. In $p+\text{Pb}$ collisions at 8.16 TeV the colliding beams have different energies per nucleon (6.5 TeV protons colliding with $(Z/A) \times 6.5 \text{ TeV} \simeq 2.6 \text{ TeV}$ per nucleon in a lead nucleus), resulting in a collision system with $\sqrt{s_{\text{NN}}} = 8.16$ TeV per nucleon pair. Under these conditions, the center-of-mass frame is shifted in rapidity, y , by $\Delta y = 0.465$ in the direction of the proton beam. Data were collected in two different periods, characterized by opposite beam orientations. Due to technical issues with the ZDC instrumentation (see Section 3.3), only **Period B** was used for this measurement.

The runs in this period were characterized by a total integrated luminosity of 55.9 nb^{-1} . Beam 1 consisted of protons and Beam 2 of Pb ions. The protons traveled toward negative rapidities in the “A→C” direction, corresponding to clockwise motion in the accelerator according to ATLAS conventions. All data sets used in this analysis, listed in Table 2, were processed using ATHENA Release 21.

2016 $\sqrt{s_{\text{NN}}} = 8.16$ TeV Data Samples	Luminosity [nb ⁻¹]
data16_hip8TeV.00313063.physics_Main.merge.AOD.r11416_p3868	0.021
data16_hip8TeV.00313067.physics_Main.merge.AOD.r11416_p3868	0.486
data16_hip8TeV.00313100.physics_Main.merge.AOD.r11416_p3868	8.787
data16_hip8TeV.00313107.physics_Main.merge.AOD.r11416_p3868	11.053
data16_hip8TeV.00313136.physics_Main.merge.AOD.r11416_p3868	9.602
data16_hip8TeV.00313187.physics_Main.merge.AOD.r11416_p3868	3.030
data16_hip8TeV.00313259.physics_Main.merge.AOD.r11416_p3868	4.659
data16_hip8TeV.00313285.physics_Main.merge.AOD.r11416_p3868	4.272
data16_hip8TeV.00313295.physics_Main.merge.AOD.r11416_p3868	9.923
data16_hip8TeV.00313333.physics_Main.merge.AOD.r11416_p3868	3.791
data16_hip8TeV.00313435.physics_Main.merge.AOD.r11416_p3868	0.301

Table 2: Data samples from $\sqrt{s_{\text{NN}}} = 8.16$ TeV p +Pb collisions collected during the 2016 Heavy Ion Run. The luminosity collected for each run is reported. All the data were processed using ATHENA Release 21.

4.2.2 Trigger Configuration

Jet triggers used for Period B of the 2016 p +Pb run are listed in Table 5. All data comes from the physics_Main stream. The working points of the triggers are described further in Section 4.3.

4.2.3 Monte Carlo Samples

In order to evaluate the performance of jet reconstruction and the jet response within ATLAS, dijet Monte Carlo samples generated using PYTHIA8 [26] with the A14 tune and the NNPDF23LO PDFs [27] were produced. Dijet events are filtered using the JZ*R04 filters, which select events containing an anti- k_t $R = 0.4$ jet in specific truth p_T ranges, referred to using the prefix “JZ” and an integer ranging from 1–5 in this analysis.

The detector response was simulated using the full ATLAS detector simulation based on the Geant4 toolkit [28]. For p +Pb collisions, Pythia8 truth events are overlaid with real minimum-bias p +Pb data. For each event, the Pythia generation and subsequent simulations were run with conditions matching the collected data, allowing the simulation to account for underlying-event effects in the jet reconstruction. In this MC

sample, the event activity originates mainly from the overlay contribution and therefore has no associated truth information.

Tables 3 and 4 list the p_T range, Pythia8 cross section and filter efficiency, and the number of events for the inclusive and forward-filtered p +Pb MC samples. For forward-filtered samples, a specific filter characterized by its own efficiency is applied to preferentially select events in which a jet exists at forward η , where the cross section is lower than for central jets. This allows smaller MC productions to provide increased statistics for events characterized by dijets in the forward region, namely low x_{Pb} and high x_p , providing satisfactory MC population over the full phase space considered. For the forward-filtered samples generated for this analysis, only jets in the proton-going direction are considered forward.

JZ	p_T^{truth} [GeV]	σ [nb]	ϵ	N_{evt}
1	20–60	7.3070×10^7	5.8782×10^{-3}	4M
2	60–160	1.2916×10^6	5.6911×10^{-3}	4M
3	160–400	1.1775×10^4	7.4572×10^{-3}	4M
4	400–800	8.9941×10^1	8.1320×10^{-3}	4M
5	>800	1.1951	7.1577×10^{-3}	4M

Table 3: Summary of inclusive $\sqrt{s_{\text{NN}}} = 8.16$ TeV p +Pb Pythia8 MC samples.

JZ	p_T^{truth} [GeV]	σ [nb]	ϵ	N_{evt}
1	20–60	7.3070×10^7	9.0671×10^{-4}	4M
2	60–160	1.2916×10^6	5.3628×10^{-4}	2M

Table 4: Summary of forward-filtered $\sqrt{s_{\text{NN}}} = 8.16$ TeV p +Pb Pythia8 MC samples.

The ATLAS data-set names for the MC samples used in this analysis are listed below.

$\sqrt{s_{\text{NN}}} = 8.16$ TeV p +Pb Inclusive

```
mc16_pPb8TeV.420011.Pythia8EvtGen_A14NNPDF23L0_jetjet_JZ1R04.merge.A0D.e6518_d1603*
mc16_pPb8TeV.420012.Pythia8EvtGen_A14NNPDF23L0_jetjet_JZ2R04.merge.A0D.e6394_d1603*
mc16_pPb8TeV.420013.Pythia8EvtGen_A14NNPDF23L0_jetjet_JZ3R04.merge.A0D.e6518_d1603*
mc16_pPb8TeV.420014.Pythia8EvtGen_A14NNPDF23L0_jetjet_JZ4R04.merge.A0D.e6518_d1603*
```

mc16_pPb8TeV.420015.Pythia8EvtGen_A14NNPDF23L0_jetjet_JZ5R04.merge.AOD.e6518_d1603*

$$\sqrt{s_{\text{NN}}} = 8.16 \text{ TeV } p\text{+Pb Forward-filtered}$$

mc16_pPb8TeV.420018.Pythia8EvtGen_A14NNPDF23L0_jetjet_JZ1R04_MaxEta_m3p0.merge.AOD.e8288_d1603*

mc16_pPb8TeV.420019.Pythia8EvtGen_A14NNPDF23L0_jetjet_JZ2R04_MaxEta_m3p0.merge.AOD.e8288_d1603*

4.3 Event Selection and Jet Reconstruction

4.3.1 Event Selection

To be included in the analysis, events must pass the relevant good run list (GRL) selection. A new GRL for the 2016 8.16 TeV $p\text{+Pb}$ data was created for this analysis to remove periods where the ZDC had an intolerable defect. This GRL can be found at:

GoodRunsLists/data16_hip/20240611/data16_hip8TeV.periodAllYear_HEAD_Unknown_PHYS_HeavyIonP_All_Good_WITH_IGNORES.xml.

4.3.2 Pileup Rejection

To have a meaningful determination of per-event centrality, it is essential to analyze events where only one hadronic $p\text{+Pb}$ collision occurred. To meet this requirement, an effective pile-up rejection strategy is needed, in particular for data with high μ (by Heavy Ion standards) like those analyzed in this note (μ ranging from 0.15 to 0.3 in most of the runs). The same pile-up rejection strategy described in [29], specifically established for this data-set, is chosen for this analysis. Events characterized by more than one vertex with 7 or more associated tracks are rejected at the level of event selection. By applying this pile-up rejection, residual pile-up contributions can be still noticed in the ZDC, see Figure 10 where the correlation between $E_{\text{ZDC}}^{\text{Pb-side}}$ and E_{T}^{Pb} is demonstrated.

This residual pile-up is not seen in the E_{T}^{Pb} spectrum, indicating that these backgrounds come from interactions that either only produce particles at very far forward rapidities or happen away from the IP. This residual was found to be present even when removing all the secondary vertices, and is suspected to be related to UPC $p\text{+Pb}$ events.

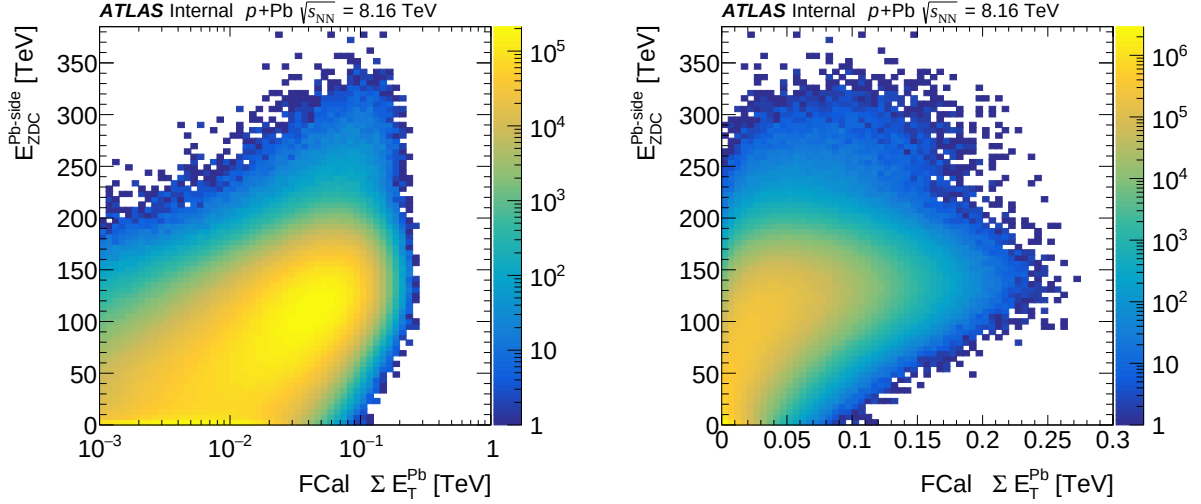


Figure 10: Correlation between the Pb going neutral forward energy ($E_{\text{ZDC}}^{\text{Pb-side}}$) and the transverse energy accumulated in the Pb-going FCal arm (E_{T}^{Pb}) for the $p+\text{Pb}$ orientation. E_{T}^{Pb} is binned in logarithmic (left) and linear (right) fashion. $E_{\text{ZDC}}^{\text{Pb-side}}$ is normalized to the single neutron peak in order to display energy in terms of number of neutrons.

A procedure to characterize and remove this additional pile-up from the ZDC distributions is discussed in Section

ATLAS Liquid Argon calorimeter [30] cells receive ionization pulses resultant from the passage of charged particle tracks. These analog pulses are delivered to a Front-End Board (FEB) where they are amplified and shaped. The characteristic shape of these FEB pulses is a large positive amplitude pulse lasting around 100 ns proceeded immediately by a smaller negative amplitude pulse lasting around 400 ns. In the 2016 8.16 TeV $p+\text{Pb}$ dataset, bunches of colliding particles circulate with 100 ns spacing between them. The measurement of E_{T}^{Pb} is then biased negatively in events with preceding bunches having large event activity. Due to the physical constraints of transmission lines and the ZDCs location far from the interaction point, pulses seen in the ZDC have long characteristic decay time. Therefore, by sampling the ZDCs ADC value just before a pulse arrives and comparing that to a common baseline it is possible to identify these events with preceding large event activity.

Due to the difference in Z between the Pb ion (82) and the proton (1), most of the

events from UPC p +Pb collisions are expected to involve the emission of a virtual photon from the Pb ion to the p , with subsequent low activity on the Pb-going side of the FCal. By applying a 0–90% selection for the per-event dijet yield analysis, one can expect a substantial fraction of these events to be effectively rejected. Diffractive events are associated with significant rapidity gaps in the particle production. To further enforce the UPC events rejection, a cut based on rapidity gaps, equal to one utilized in the centrality analysis [29], where events with $\Delta\eta_{\text{gap}}^{\text{Pb}} > 1.8$ were rejected. In this analysis, such a cut rejects only a handful of events in the 0-90% centrality class, as demonstrated by Figure [ADD Gaps figure](#).

4.3.3 Jet Reconstruction

The jet reconstruction is common to both measurements reported in this thesis and is detailed in Section [3.4](#).

4.3.4 Jet Calibration

For heavy ion jets in the 2016 dataset¹⁰, the in-situ portion of the jet calibration was taken from **EMTopoJets**¹¹. A cross-calibration, derived using the direct balance method in events containing a Z boson or photon opposite to a jet, is used to bridge between **AntiKt4HIJets** and fully calibrated **EMTopoJets** jets by accounting for additional differences in response between them [23]. Thanks to the cross-calibration, it is possible to characterize the performance of the **AntiKt4HIJets** without deriving a whole new in-situ calibration. This procedure also has an associated systematic uncertainty.

¹⁰For an overview of the heavy ion jet calibration, see section [3.4.2](#). This section will only describe peculiarities of the calibration specific to the 2016 dataset.

¹¹**EMTopoJets** are calorimeter jets reconstructed somewhat similarly to **AntiKt4HIJets** but without any UE subtraction. The **EMTopoJets** are calibrated by the ATLAS JetEtMiss group which has much more personpower than the ATLAS heavy ion jet group.

4.3.5 Jet Selection

Jets are reconstructed using the anti- k_t jet algorithm with a radius parameter of $R = 0.4$.

Jets are required to pass the ‘LooseBad’ jet selection working point.

4.3.6 Jet Trigger Strategy

Events are considered only if the leading jet in the trigger has fired the corresponding trigger chosen for the (η, p_T) region where the jet is reconstructed. The trigger selection for $|\eta| < 3.0$ and for $|\eta| > 3.5$ above a p_T of 40 GeV is straightforward. In these regions, the fully efficient trigger associated with the lowest prescale value is chosen.

Efficiency values used to determine what trigger to choose in each (η, p_T) region are reported in Table 5. For triggers covering the same η region, prescale values decrease as the p_T threshold increases.

A different treatment is requested in the η region corresponding to the transition between central and forward triggers, i.e. around $\eta = \pm 3.2$ ¹². To handle the trigger analysis in this region, an inclusive approach (see [31] and references therein), hereafter referred to as the Fully Efficient Trigger Combination (FETC) method, was chosen. In the FETC approach, a combined weight based on all the considered trigger items for the given kinematic region is determined for the entire event sample. For each event, at least one actual trigger item bit from the set of considered items is required to be set. The weight ($w_{j,k}$) calculation for an event j in a given run k is based on the probability to accept the event after the application of the prescale to all the i -th trigger items considered. If $r_{i,j}$ is the bit for the trigger item i in the event j , and the same trigger item is prescaled by a factor $p_{i,k}$ during the run k , the probability that the event j in the

¹²Sign here is determined by the beam orientation, since this analysis covers only central and forward regions, discarding the backward where the energy accumulated in the FCal is used for $\langle \text{FCal} \sum E_T^{\text{Pb-side}} \rangle$.

run k is accepted by the trigger item i after the prescale can be expressed as

$$q_{i,j,k} = \frac{r_{i,j}}{p_{i,k}} \quad (17)$$

Assuming the prescale decisions for each trigger item are independent from each other, the probability that at least one of the N_{items} trigger items accepts the event is given by

$$q_{j,k} = 1 - \prod_{i=1}^{N_{\text{items}}} \left(1 - \frac{r_{i,j}}{p_{i,k}} \right) \quad (18)$$

Starting from this, one can define the run dependent weight for event j as

$$w_{j,k} = \frac{1}{q_{j,k}} \quad (19)$$

When combining all the considerations outlined above, one obtains a (η, p_T) map that provides, per each jet with $p_T > 40$ GeV and $-4.5 < \eta < 3.1$ ($-3.1 < \eta < 4.5$), the jet trigger(s) considered for the analysis of the $p+\text{Pb}$ ($\text{Pb}+p$) data. The three maps corresponding to the trigger configurations used in the $p+\text{Pb}$ run are shown in Figure [add trigger map](#).

The analysis reported in this note will use dijet events with a leading jet $p_T > 40$ GeV and a sub-leading jet $p_T > 30$ GeV. Only the dataset using $p+\text{Pb}$ -ion trigger scheme is considered for the analysis presented in the CONF note, while the other two periods are used just for studies included in the Appendix of this Note and reported here for completeness.

4.4 Analysis Procedure

The Pb-going FCal transverse energy is constructed by summing the transverse projection of the electromagnetic-scale tower energies in the hemisphere facing the Pb beam. This definition is consistent with previous ATLAS $p+\text{Pb}$ centrality measurements.

Trigger Chain	p_T^{99} [GeV]	η	$\mathcal{L}$ [ub ⁻¹]	p_T^{99} Monitor
HLT_j30_ion_0eta490_L1TE10	36	[−4.9,4.9]	113.76	HLT_mb_sptrk
HLT_j40_ion_L1J5	44	[−3.2,3.2]	468.1	HLT_mb_sptrk
HLT_j50_ion_L1J10	57	[−3.2,3.2]	1120	HLT_mb_sptrk
HLT_j60_ion_L1J20	70	[−3.2,3.2]	1562	HLT_mb_sptrk
HLT_j75_ion_L1J20	80	[−3.2,3.2]	3468	HLT_mb_sptrk
HLT_j100_ion_L1J20	111	[−3.2,3.2]	56671	HLT_mb_sptrk
HLT_j15_ion_n320eta490_L1MBTS_1_1	44	[−3.2,−4.9]	77.5	HLT_mb_sptrk
HLT_j25_ion_n320eta490_L1TE5	40	[−3.2,−4.9]	247.7	HLT_mb_sptrk
HLT_j35_ion_n320eta490_L1TE10	57	[−3.2,−4.9]	179.9	HLT_j25_ion_n320eta490_L1TE5
HLT_j45_ion_n320eta490	65	[−3.2,−4.9]	5288	HLT_j25_ion_n320eta490_L1TE5
HLT_j55_ion_n320eta490	80	[−3.2,−4.9]	16635	HLT_j25_ion_n320eta490_L1TE5
HLT_j65_ion_n320eta490	85	[−3.2,−4.9]	53571	HLT_j25_ion_n320eta490_L1TE5

Table 5: Overview of p +Pb ion triggers used in the analysis. The sampled luminosity, $\mathcal{L}$, recorded by each trigger is reported. The L1 for triggers that do not specify this detail in the chain name are: L1_J15.31ETA49 for HLT_j45_ion_n320eta490 and HLT_j55_ion_n320eta490, and L1_J20.31ETA49 for HLT_j65_ion_n320eta490. p_T^{99} represents the 99% efficiency point for a given chain, calculated taking into account both L1 and HLT efficiencies, and is reported in GeV.

In typical events containing a hard scattering, the energy observed in the Pb-going ZDC is dominated by spectator neutrons evaporating from the struck nucleus.

The ZDC energy scale is determined run by run from the response to a single spectator neutron. The low-energy spectrum contains distinguishable neutron peaks, and a simultaneous fit to the first three peaks and a background term validates the calibration. The fitted single-neutron peak is constrained to the nominal per-nucleon beam energy. The detector response below 1 TeV is limited by detector noise and low-energy backgrounds; this region is treated as a dedicated zero-energy bin. A residual in-time pileup contribution to the $E_{\text{ZDC}}^{\text{Pb-side}}$ distribution is removed with a data-driven template method.

Finite jet-resolution effects cause migration in x_p . These effects are corrected with a two-dimensional iterative Bayesian unfolding procedure using RooUnfold. The response matrix contains x_p and, in turn, either $E_{\text{ZDC}}^{\text{Pb-side}}$ or E_T^{Pb} . Since the available simulation does not model nuclear breakup in the ZDC or the full FCal response for the overlaid

minimum-bias event, the calorimeter observable is propagated through the unfolding with diagonal migration matrices. This retains the measured correlation of calorimeter energy with x_p while correcting the jet-related kinematic migration.

The response matrix is reweighted as a function of reconstructed x_p to match the data spectrum. For the ZDC measurement, pileup-subtracted data distributions are sampled to assign a ZDC energy to each simulated dijet event. The procedure is repeated with independent random seeds, and closure tests verify that the unfolding reproduces the particle-level simulation. Two iterations are used, balancing the convergence of the unfolded result against the growth of statistical uncertainty. Statistical uncertainties are evaluated by bootstrap resampling of both the data and response matrix.

4.5 Systematic Uncertainties

Systematic uncertainties arise from the jet energy scale (JES), jet energy resolution (JER), unfolding prior, disabled HEC sector, and the ZDC energy calibration. Except for the HEC and ZDC-scale variations, the full analysis is repeated with a response matrix modified according to each systematic variation. The difference relative to the nominal result defines the corresponding uncertainty.

The JES and JER variations quantify the mean and width of the $p_T^{\text{reco}}/p_T^{\text{gen}}$ response for matched jets. The HEC uncertainty is evaluated by enlarging the jet-exclusion region by 0.1 in both pseudorapidity and azimuth. To test the sensitivity to the unfolding prior, the response matrices are constructed without the nominal event-level reweighting and the full unfolding is repeated.

The ZDC-specific uncertainty includes the stability of the single-neutron peak, the residual energy-scale variation, and the non-linear energy correction. The FCal stability across the data-taking period is negligible relative to the other sources. The independent variations are added in quadrature. For ratios between x_p intervals, JES, JER, HEC, and ZDC uncertainties are treated as fully correlated, while the unfolding-prior contribution

is treated as uncorrelated.

For the normalized distributions, the $E_{\text{ZDC}}^{\text{Pb-side}}$ uncertainty is typically dominated by the ZDC energy-stability term and is below 7% over most of the measured range. The E_{T}^{Pb} distribution uncertainty is primarily driven by JES, JER, and the unfolding prior and is generally below 10%. The systematic uncertainties on $\langle E_{\text{ZDC}}^{\text{Pb-side}} \rangle$ and $\langle \text{FCal} \sum E_{\text{T}}^{\text{Pb-side}} \rangle$ are below 2% in all but the lowest x_p interval.

4.6 Results

The normalized E_{T}^{Pb} and $E_{\text{ZDC}}^{\text{Pb-side}}$ distributions change systematically with x_p . Relative to a low- x_p selection, high- x_p dijet events exhibit an E_{T}^{Pb} distribution shifted toward lower energies. This is a direct confirmation of the event-activity bias previously observed in centrality-dependent jet measurements. The $E_{\text{ZDC}}^{\text{Pb-side}}$ distribution shows a qualitatively similar trend: at high x_p , a larger fraction of events contains fewer forward neutrons. The effect in the ZDC is, however, substantially weaker than that in the FCal.

The mean values $\langle E_{\text{ZDC}}^{\text{Pb-side}} \rangle$ and $\langle \text{FCal} \sum E_{\text{T}}^{\text{Pb-side}} \rangle$ both decrease with increasing x_p , most visibly for $x_p \gtrsim 0.02$, where earlier ATLAS measurements found evidence for small proton-size configurations. Across this range, $\langle E_{\text{ZDC}}^{\text{Pb-side}} \rangle$ decreases by up to about 5%, corresponding on average to two fewer beam-energy neutrons, whereas $\langle \text{FCal} \sum E_{\text{T}}^{\text{Pb-side}} \rangle$ decreases by up to 40%. When the two means are normalized to their maximum values, the FCal estimator is found to be approximately six times more sensitive to hard-process kinematics than the ZDC estimator.

The relation between E_{T}^{Pb} and $E_{\text{ZDC}}^{\text{Pb-side}}$ is also x_p dependent. For ZDC energies below approximately 150 TeV, the average FCal transverse energy is consistent with a linear dependence on the ZDC energy in low-, middle-, and high- x_p selections. The slope decreases progressively from low to high x_p , and the Pearson correlation coefficient changes by approximately 3% across the measured range. At higher ZDC energies, deviations from the linear trend are consistent with larger fluctuations in nuclear breakup

for events with the greatest event activity.

These results show that both common geometry estimators retain sensitivity to the proton configuration associated with a hard scattering. The ZDC observable is more robust than Pb-going FCal transverse energy, but is not independent of the hard-scattering kinematics. This observation provides experimental input for simultaneous descriptions of event activity, nuclear breakup, and color fluctuations in p +Pb collisions.

Chapter 5 Observation of centrality-dependent dijet transverse momentum imbalance in O+O and Ne+Ne collisions

5.1 observable

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5.2 Data Sets

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5.3 Event Selection and Jet Reconstruction

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5.3.1 Event Selection

5.3.2 Pileup Rejection

5.3.3 Jet Reconstruction

The jet reconstruction is common to both measurements reported in this thesis and is detailed in Section [3.4](#).

5.3.4 Jet Calibration

In Run 3, the in-situ and η -intercalibration¹³ calibrations were investigated in detail for `AntiKt4HIJets` for the first time using high- μ pp data at 13.6 TeV, leveraging the independence of the calibration on $\sqrt{s_{\text{NN}}}$. The η -intercalibration is performed using the matrix-minimization method described in [\[32\]](#), with the central reference method used as a cross check. Calibration factors are derived in bins of p_{T} and η and are applied as a correction to the jet p_{T} . Cross-checks of the calibration have been established using

¹³In Run 3 the ATLAS JetEtMiss group stopped supporting the `EMTopoJets` so the cross calibration strategy used in Run 2 was not an option.

dedicated MC samples at $\sqrt{s_{\text{NN}}}= 5.36$ TeV and residual non-closure resulted in additional systematic uncertainty. The work also included studies of the systematic uncertainty associated with each component of the calibration, as well as the jet energy resolution (JER).

5.3.5 Jet Selection

5.3.6 Jet Trigger Strategy

5.4 analysis

Placeholder text for this section.

5.5 uncertainties

Placeholder text for this section.

5.6 results

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Chapter 6 Summary and Conclusions

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